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ON SYSTEMS WITH TRANSFER FUNCTIONS RELATED TO THE RIEMANN ZETA FUNCTION

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INTRODUCTION.

The qualitative theory of differential equations has been successful in solving many fundamental problems of the theory of control. The development of systems which contain computers leads to new types of problems which require special investigation. Such systems can operate in a succession of nonidentical steps of increasing complexity. Moreover, some critical quantities in the system (e.g. the feedback function) can be determined by testing whether a certain condition is satisfied and by choosing the value of the quantity depending on the result of the test ("logical feedback"). At this level of complexity, new types of behavior may arise at any step and the problem of predicting the future behavior of the systems becomes particularly acute. Sometimes it is not even clear if, or to what extent, the long range behavior is predictable. In any case, although these applications deal with purely deterministic processes executed by machines, the machines themselves are unable to predict their future behavior. It is a task of the theory to study mathematical models of these systems in order to elucidate their behavior.

In this paper, we present three of the simplest mathematical models related to our subject. The first two are simple interpretations, in the control theory framework, of known results of Möbius and von Mangoldt. Both of these systems have transfer functions related to Riemann's zeta function. The qualitative behavior of the systems depends on the location of the zeros of the function zeta, a problem which is still in the conjecture-checking stage. Both models illustrate very clearly the difficulties mentioned above. It is perhaps natural that, in trying to understand the behavior of systems of increasing complexity, one is faced with mathematical problems of extreme difficulty.

Both Möbius and von Mangoldt models are purely algebraic. The third model given in this paper is a nonlinear dynamical system with unbounded delays. This system is studied in more detail, by combining some standard results in the analytic number theory with the frequency domain approach of control theory.

A MATHEMATICAL MODEL FOR A PURE COMPUTER FEEDBACK SYSTEM (MÖBIUS SYSTEM).

Let u be a real function, defined for $T > 1$. We define the Möbius system as

$$x(T) = -\sum_{n=2}^{\infty} x\left(\frac{T}{n}\right) + u(T) \quad , \text{ for } \quad T > 1, \tag{1}$$

$$x(T) = 0 \quad \text{if } 0 < T \leq 1 \tag{2}$$

The "solution" is the unique function x, defined for $T > 1$, which satisfies (1) and (2). These equations can be interpreted as describing a feedback system (fig. 1).

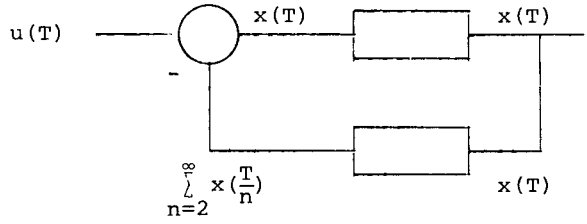


fig. 1

Notice that if u is integer-valued then so also is the solution x. Hence the problem can be handled by computers with absolute accuracy.

From the viewpoint of the qualitative theory of systems, one meets the following problem: Given u (e.g. $u = 1$) find precise estimates of x. But although x can be written explicitly (by Möbius inversion formula) the problem remains unsolved. The theory leads to some (upper and lower) estimates, between which there is an enormous gap. Nor can this problem be solved by running it on a computer (in spite of the fact that it seems to be ideally suited for such an approach). In fact the computer is able to determine precisely the solution on finite intervals, but is absolutely useless for finding long range estimates. Similar comments remain valid for the other systems considered below.

In order to apply traditional methods of control theory, it is convenient to put $t = \log T$ and to consider, instead of (1),(2), the system

$$x(t) = -\sum_{n=2}^{\infty} x(t-\log n) + u(t) \quad , \text{ if } t > 0$$
$$x(t) = 0, \quad \text{if} \quad t \leq 0,$$

the solution x being now defined on the whole real axis R. It is easy to see that the transfer function of this system (from u to x) is equal to

$$1/\sum_{n=1}^{\infty} n^{-s} \quad , \text{ for } \mathrm{Re}\, s > 1,$$

i.e. it equals the inverse of Riemann's zeta function, $\zeta(s)$.

The qualitative behavior of this system depends on the location of the zeros of the function zeta - and the elucidation of this difficult problem constitutes the only way for understanding the long range behavior of this system, in spite of its apparent simplicity.

A MATHEMATICAL MODEL FOR SYSTEMS WITH LOGICAL FEEDBACK (VON MANGOLDT SYSTEM).

Let $\Lambda(n)$ be von Mangoldt's function (equal to $\log p$ if n is equal to a prime p or to a power of the prime p and equal to zero otherwise). We define the von Mangoldt's feedback system by

$$\sum_{n=2}^{\infty} \Lambda(n)\, x(t-\log n) = u(t), \text{ if } t > 0 \ ,$$

$$x(t) = 0 \ , \text{ if } t \leqq 0.$$

This system can again be interpreted as a feedback system, but the coefficients $\Lambda(n)$ must be determined by a test (checking whether n is a power of a prime). Hence this model belongs to the category of systems with logical feedback.

The inverse of the transfer function of this system is $\sum_{n=2}^{\infty} \Lambda(n) n^{-s}$, if Re s > 1. It is known that

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty} \Lambda(n) n^{-s} \ , \text{ if Re } s > 1 \tag{3}$$

[1,(1.1.8) p.4]. Moreover

$$-\frac{\zeta'(s)}{\zeta(s)} = -\sum_{\rho} \frac{1}{s-\rho} + 1 - \sum_{\rho} \frac{1}{\rho} - \frac{\zeta'(0)}{\zeta(0)} + \frac{1}{s-1} + \sum_{n=1}^{\infty} \frac{s}{2n(s+2n)} \tag{4}$$

where ρ runs over all the nonreal zeros of $\zeta(s)$ [2,p.52]. It is known that all the nonreal zeros ρ satisfy the condition Re $\rho \in (0,1)$ (they lie in the "critical strip", symmetrically about the point (1/2,0)). The qualitative behavior of the solution of von Mangoldt's system depends on the upper bound of Re ρ - which will be denoted in the following by Θ. But all one knows is that Θ lies somewhere in the interval [1/2,1].

A NONLINEAR DYNAMICAL SYSTEM WITH LOGICAL FEEDBACK.

One can now study a new class of feedback systems, introducing another feedback law, namely assuming that u is related to x according to some suitable feedback relation (this adds a new feedback loop in fig. 1, but the new loop can be combined with the old one to obtain a single feedback loop). The behavior of the solutions will depend, of course, in an essential way on the new feedback relation. It is an interesting problem to find feedback relations for which the growth of the solution is determined by the quantity Θ. The next model satisfies this condition. Moreover the feedback law is nonlinear and contains the derivative of the solution, such that the result is a dynamical system whose form is not very exotic. The system is described by the equations

$$\sum_{n=2}^{\infty} \Lambda(n)\, x(t-\log n) - \int_{-\infty}^{t} e^{t-\tau} x(\tau)\, d\tau - \int_{0}^{t} \sum_{n=1}^{\infty} \frac{1}{2n} e^{-2n(t-\tau)} \dot{x}(\tau)\, d\tau = \dot{x}(t) + f(x(t)), \text{ if } t > 0 \tag{5}$$

$$x(t) = \phi(t), \text{ if } t \leqq 0, \tag{6}$$

where ϕ is a C^1 function with compact support, defined for $t \leqq 0$ (relation (6) defines an initial condition). The function f is assumed to be globally Lipschitzian, from R into R. By "solution" of (5),(6) we mean a real C function x, defined for $t \in R$, such that the series in (5) converge and relations (5) and (6) are satisfied.

Since the coefficients $\Lambda(n)$ are determined by a test, as in the von Mangoldt case, the model can be interpreted as a system with logical feedback.

The system falls into the category of the dynamical systems with unbounded delays [3]. A similar system has been considered in [4]. The fact that the kernel in the last integral from (5) is not bounded makes the problem somewhat peculiar, but a contraction-mapping argument (which we omit) proves that, for every C^1 function ϕ with compact support, there exists a unique C solution x of (5), (6).

This model is just a representative of a huge class of systems which can be treated similarly.

In order to estimate the solution x we introduce the notation

$$|\phi| = \sup_{t \leqq 0} (e^{-\theta t}|\phi(t)|) + \int_{-\infty}^{0} e^{-\theta t}|\phi'(t)|dt,$$

where θ is as in the following statement:

Theorem. Suppose that the globally Lipschitzian function f satisfies the condition $f(r)r \geqq 0$, for every real r. Then for every $\theta > \Theta$ there exists a constant k such that, for every C^1 function ϕ with compact support, the solution x of (5),(6) satisfies the inequality

$$|x(t)| \leqq k\, e^{\theta t}|\phi|, \text{ for every } t > 0.$$

The proof uses some standard results of the theory of distribution of prime numbers (see e.g. [5]) and the main ideas of the frequency domain method (see e.g. [6]-[8]).

Fix the initial condition ϕ and the solution x of (5),(6). For each $T > 0$, define the functions y_T, w_T and z_T, from $(0,\infty)$ into R, as follows

$$y_T(t) = \begin{cases} e^{-\theta t}x(t) & \text{if } 0 < t \leqq T \\ 0 & \text{if } t > T \end{cases}$$

$$w_T(t) = \begin{cases} e^{-\theta t}\dot{x}(t) & \text{if } 0 < t \leqq T \\ 0 & \text{if } t > T \end{cases}$$

$$z_T(t) = \sum_{2 \leqq n < e^t} \Lambda(n) n^{-\theta} y_T(t - \log n) - \int_0^t e^{(1-\theta)(t-\tau)} y_T(\tau)\, d\tau$$

$$- \sum_{n=1}^{\infty} \frac{1}{2n} \int_0^t e^{-(2n+\theta)(t-\tau)} w_T(\tau)\, d\tau$$

$$- x(0) \sum_{n=1}^{\infty} \frac{1}{2n} e^{-(2n+\theta)t} + e^{-\theta T} x(T) S_T(t), \qquad (7)$$

where

$$S_T(t) = \begin{cases} 0 & \text{if } 0 < t \leqq T \\ \sum_{n=1}^{\infty} \frac{1}{2n} e^{-(2n+\theta)(t-T)} & \text{if } t > T. \end{cases}$$

A space-consuming computation shows that z_T tends exponentially to

zero as t tends to infinity. (Since the main idea of this computation is contained in the proof of a Lemma below, we omit the details) It follows that the Laplace transform

$$\hat{z}_T(s) = \int_0^\infty e^{-st} z_T(t)\,dt$$

is analytic for Re $s > -\varepsilon$, where ε is some strictly positive number. Notice also that the Laplace transform $\hat{y}_T(s)$ is entire. Using (5) it is easy to see that, if $0 < t < T$, then

$$z_T(t) = e^{-\theta t}(\dot{x}(t) + f(x(t)) - \sum_{n \geq e^t} \Lambda(n)\phi(t-\log n) - x(0)\sum_{n=1}^{\infty} \frac{1}{2n} e^{-2nt} + \int_{-\infty}^{0} e^{t-\tau}\phi(\tau)\,d\tau), \quad 0<t<T, \quad (8)$$

since $S_T(t) = 0$ in the interval $0 < t < T$.

From (7) one finds that, if Re $s + \theta > 1$, then

$$\hat{z}_T(s) = \left(\sum_{n=2}^{\infty} \Lambda(n) n^{-(s+\theta)} - \frac{1}{s+\theta-1} - \sum_{n=1}^{\infty} \frac{s+\theta}{2n(s+\theta+2n)} \right) \hat{y}_T(s).$$

Moreover from (3) and (4) it follows that, if Re $s + \theta > 1$, then

$$\hat{z}_T(s) = \left(c - \sum_{\rho} \frac{1}{s+\theta-\rho} \right) \hat{y}_T(s), \quad (9)$$

where

$$c = 1 - \frac{\zeta'(0)}{\zeta(0)} - \sum_{\rho} \frac{1}{\rho}.$$

However, since the functions involved in (9) are analytic for Re $s > -\varepsilon$, the relation is also true for Re $s = 0$. From the known relations $\zeta'(0)/\zeta(0) = \log 2\pi$ [2,p.66] and

$$\sum_{\rho} \frac{1}{\rho} = 1 + \gamma/2 - (\log \pi)/2 - \log 2$$

[2, pp. 159,160] - where γ is Euler's constant - it follows that $c < 0$. Moreover, our assumption $\theta > \text{Re } \rho$ implies that

$$\text{Re } (1/(i\omega+\theta-\rho)) > 0, \text{ for every real } \omega.$$

These results allow us to obtain, from (9), the relation

$$\hat{z}_T(i\omega) = G(i\omega)\hat{y}_T(i\omega), \text{ for every real } \omega,$$

where Re $G(i\omega) < 0$, for every real ω. Hence, if one defines

$$J_T = \int_0^\infty y_T(t)\, z_T(t)\, dt$$

and one applies Parseval formula (which is legitimate) one obtains

$$J_T \leq 0. \quad (10)$$

On the other hand, from (8), we get

$$J_T = \int_0^T e^{-2\theta t} x(t)(\dot{x}(t) + f(x(t)) - x(0)\sum_{n=1}^{\infty} \frac{1}{2n} e^{-2nt} - \sum_{n > e^t} \Lambda(n)\phi(t-\log n) + \int_{-\infty}^{0} e^{t-\tau}\phi(\tau)\,d\tau)\, dt.$$

Part of this expression can be estimated as in the following lemma (whose proof is given in the next section).

Lemma. There exists a constant K (independent of T) such that

$$\left| \int_0^T e^{-2\theta t} x(t) \Big(\sum_{n > e^t} \Lambda(n)\phi(t-\log n) - \int_{-\infty}^{0} e^{t-\tau}\phi(\tau)d\tau \Big)\, dt \right|$$

$$\leq K|\phi| \sup_{0<\tau<T} \left(e^{-\theta\tau}|x(\tau)|\right), \text{ for every } T > 0.$$

From this lemma and from the condition $f(r)r \geq 0$ it follows that (10) becomes

$$\int_0^T e^{-2\theta t} x(t)\dot{x}(t)dt \leq x(0) \int_0^T e^{-2\theta t} x(t) \sum_{n=1}^{\infty} \frac{1}{2n} e^{-2nt} dt$$

$$+ K|\phi| \sup_{0<\tau<T} (e^{-\theta\tau}|x(\tau)|) \qquad (11)$$

The left hand side can be written as

$$\frac{1}{2} \int_0^T e^{-2\theta t} \frac{d}{dt} (x^2(t))dt = \frac{1}{2} e^{-2\theta T} x^2(T) - \frac{1}{2} \phi^2(0)$$

$$+ \theta \int_0^T e^{-2\theta t} x^2(t)dt$$

and the first term in the right hand side of (11) can be estimated as follows: there exists a constant k' such that

$$\int_0^T e^{-2\theta t} x(t) \sum_{n=1}^{\infty} \frac{1}{2n} e^{-2nt} x(0)dt \leq k'|\phi(0)| \sup_{0<\tau<T} (e^{-\theta\tau}|x(\tau)|)$$

Putting $y(t) = e^{-\theta t}x(t)$, inequality (11) becomes

$$y^2(T) \leq \phi^2(0) + K'|\phi| \sup_{0<\tau<T} |y(\tau)|,$$

for some new constant K'. Since the condition is valid for every $T > 0$, the quantity $v(T) = \sup_{0<\tau<T} |y(\tau)|$ satisfies the inequality

$$v^2(T) \leq \phi^2(0) + K'|\phi|\, v(T).$$

By completing the square one obtains the conclusion of the theorem.

PROOF OF THE LEMMA.

Using the function

$$\psi(x) = \sum_{n \leq x} \Lambda(n),$$

one can write the Stieltjes integral

$$\sum_{n>e^t} \Lambda(n)\phi(t-\log n) = \int_{e^t}^{\infty} \phi(t-\log x)\, d\psi(x)$$

$$= \int_{e^t}^{\infty} \phi(t-\log x)\, d(\psi(x)-x) + \int_{e^t}^{\infty} \phi(t-\log x)dx.$$

The last integral equals $\int_{-\infty}^{0} e^{t-\tau}\phi(\tau)d\tau$. The first integral in the right hand side is integrated by parts and one uses the fact that there exists a number $r < \theta$ such that

$$\psi(x) - x = O(x^r).$$

This last fact follows from our assumption on θ and from Theorem 30 in [5,p.83]. With these results, the inequality to be proved follows easily from our assumptions on the function ϕ.

CONCLUSIONS.

Problems which have been intensively studied for a long time for their own sake turn out to be relevant for understanding the behavior of some actual systems in control theory. Inherent difficulties make the theory still incomplete, in spite of more than a century of efforts and continuous progress. The fundamental interest of these problems in so many different areas points to the need for further investigations.

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This paper is in final form and no version of it will be submitted for publication elsewhere.